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Summer Internship Company: PAYSPRINT PRIVATE
LIMITED

Summer Internship Role: OPERATIONS TRAINEE

State-Driven Partner Onboarding & Lifecycle Automation System

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Project Title

- **Project Description:** The project involved designing, deploying, and documenting an automated, state-driven workflow orchestration system to streamline the end-to-end partner integration lifecycle. By leveraging an integration platform to parse incoming communications, map them to a relational tracking repository, and automatically trigger conditional actions (such as generating drafts, tracking UAT sign-offs, and managing escalation schedules), the system replaces highly manual processes with a structured, visible, and error-resistant digital framework.
- **One-Line Objective:** To architect and implement an automated email-driven workflow system using n8n and Google Sheets that orchestrates partner onboarding, UAT tracking, and live transitions while eliminating manual cross-functional handoff friction.

Problem Description

- **Manual Operational Overhead:** The partner onboarding pipeline heavily relied on human intervention to parse inbound partner requests, track technical milestones, and execute routine communications, leading to severe resource bottlenecks.
- **Lax Visibility & Tracking Mismatches:** Lifecycle tracking lacked central transparency, making it difficult to immediately identify exactly where a partner was stalled (e.g., pending UAT requirements vs. live credentials) without manual email thread audits.
- **Fragmented Follow-up & Escalation Management:** Follow-ups for unresponsive partners were performed ad-hoc, lacking a disciplined, time-bound progression model (e.g., escalating precisely at Day 3, Day 5, and Day 7+) to safeguard integration momentum.

Operational Aspect			Manual State Challenges		Impact on Business Teams	
Data Integrity			Manual data entries, row merging, column name variations.		Breaks data pipelines; causes database mismatch.	
Milestone Monitoring			No automated state changes (stage_flag tracking).		Missed target timelines for UAT and Go-Live.	
Communication			Crafting standalone emails repeatedly for every product tier.		High processing time per partner account.	

Your Approach/ Work

Technical Implementation & Core Architecture

- **Environment Architecture Setup:** Built out the foundational infrastructure by establishing a secure project on Google Cloud Platform, activating the necessary APIs (Gmail, Sheets, Drive), and setting up OAuth 2.0 user-scoped consent states.
- **Engine Core Deployment:** Configured a local environment on Windows running an independent Node.js ecosystem to host, run, and scale global automated routines via the n8n orchestration core.
- **Database Relational Design:** Architected a structured, multi-tab spreadsheet to serve as the project's transaction database, incorporating rigid schemas for raw onboarding data, partner profiles, team routing rules, and master product sign-off definitions.
- **State-Driven Workflow Engineering:** Imported, optimized, and mapped multi-stage workflows within the orchestration engine to handle targeted state changes via transactional stage_flag values:
 - *Workflow 1 (New Email & Credentials):* Auto-checks incoming mails, updates records to UAT_REQUIREMENT_SENT, and creates communication drafts.
 - *Workflow 2 (Credential Sent Checker):* Validates delivery and transitions state to UAT_CREDENTIALS_SENT.
 - *Workflow 3 (Follow-Up Tracker):* Scans client replies and executes dynamic, aging follow-up loops.

Your Approach/ Work

Timeline, Skills & MBA Learnings Applied

- **Project Timelines:**

- *Weeks 1-2:* Current-state workflow mapping, GCP integration environment provisioning, and API scoping.
- *Weeks 3-5:* Database schema architecture, automated script node wiring, and core parsing logic rules.
- *Weeks 6-7:* Multi-stage lifecycle integration testing, client response loop edge-case building, and manual override handling (Refunds/Discontinuations).
- *Week 8:* Compiling full technical SOP documentation, setting system guardrails, and executing cross-functional handoff.

- **Applied Skills & Tech Stack:** Workflow Automation Orchestration, Cloud Resource Management (GCP console), Data Schema Architecture, API Integration (OAuth 2.0 protocols), and Technical Writing.

- **Applied MBA Learnings:**

- **Operations Management & Process Analysis:** Used capacity balancing principles to identify operational friction points and replace manual steps with deterministic digital routing.
- **Information Systems Management:** Implemented entity-relationship principles by leveraging `thread_ID` as a non-clashing unique primary key to achieve row-level database granularity across data tables.

- **Strategic Document Design:** Translated complex system logic into an accessible, SOP-based format for cross-functional collaboration.

Measurable Business Outcomes & System

Impact

- **Zero-Intervention Tracking Mechanics:** Created a self-sustaining data stream where client transactions migrate cleanly from email nodes directly into structured tables, reducing human setup labor.
- **Deterministic Lifecycle States:** Replaced ambiguous milestone tracking with a definitive, state-driven model that explicitly shifts records through exact flags (UAT_REQUIREMENT_SENT → UAT_CREDENTIALS_SENT → UAT_SIGNOFF_DONE → GO_LIVE_DONE), removing guesswork from the pipeline.
- **Algorithmic Follow-Up Discipline:** Introduced an automated aging escalation schedule that shifts states precisely based on timeline thresholds, instantly resetting to baseline values the moment an active partner responds.
- **Operational Continuity Assurance:** Formulated an exhaustive deployment guide and critical operating parameters to insulate the system from runtime breaks (e.g., maintaining column string formats, enforcing strict node hierarchies, and setting manual boundaries for refunds).

Key Learnings/Recommendations

- **Project & Technical Learnings:** Mastered the assembly of cloud-based integration layer architectures, gained deep fluency in cloud security access methods (OAuth scopes and redirect paths), and realized how important precise text parsing criteria are to automation robustness.
- **Organizational Experience:** Gained firsthand experience on how critical clear internal handoff documentation is when bridging engineering goals with client-facing business teams. Learned to design failsafes for technical systems to accommodate inevitable real-world operational exceptions.
- **List of Tools Learned:** n8n Workflow Orchestrator, Google Cloud Console (APIs & Services Layer), Node.js Ecosystem, Advanced Google Workspace Schema Logic.
- **Suggestions for the PGP Committee:** Recommend adding a practical component to the core Management Information Systems (MIS) or Operations curriculum that goes beyond standard software concepts. Introducing modern, no-code/low-code business automation architectures (like n8n, Make, or Zapier) would equip future product and operations managers with the immediate toolkit needed to automate messy corporate processes.

Thank You